

Analysis Report on the Technical Homology Between zw Skeleton Network and Mainstream AIGC Technologies

Report Nature: Analysis from a Neutral, Third-Party Professional Perspective

Core Data Sources: Public materials from the Zhiwang AI Official WeChat Account, the official Ziwang Column on Cnblogs, the core paper *New Ideas for Chinese Font Design: Artificial Intelligence and Chinese Font Design*, full traceable materials from the reprinted version of the 1994 *China Grand Character Library* CD-ROM, technical archives on open-source platforms, as well as public academic achievements and engineering white papers in the global AIGC field.

Comparison Subjects:

- **Party A:** The complete technical system of the zw Skeleton Network developed by He Haiqun (also known as zw/Zhiwang/Ziwang/Tongling)
- **Party B:** The controllable generation technology system applied by the world's mainstream AIGC large model manufacturers (including core skeleton/contour constraint technologies, and related text-to-image/text-to-video multimodal generation capabilities)

Core Judgment Criteria: Dual Judgment of Duck Test + Academic Theoretical Test

Report Description: All conclusions of this report are based on publicly available and traceable evidence chains. There is no subjective targeted expression for specific manufacturers, and the report only conducts general technical homology analysis, compliance risk sorting and industry recommendations.

I. Core Definitions and Benchmark for Comparison

1.1 Official Definition of Core Technologies

Technical System	Official Core Definition	Core Implementation Time	Core Application Scenarios
zw Skeleton	The world's first	R&D initiated in	AI generation of

Network	industrially implemented technical solution for "in-depth rigid fusion of skeleton and contour". Its core is to build full-process rigid constraints for the generation process through skeleton/contour extraction of the target object, so as to solve the problems of structural deformation and lack of consistency in AI-generated content.	1990, commercialized and officially named open-source in 1994	Chinese character libraries, 2D/3D image processing, and later expanded to general AIGC full scenarios
Controllable Generation Capabilities of Mainstream AIGC Large Models	The core text-to-image and text-to-video multimodal capabilities of large models under major manufacturers, natively integrated with a controllable generation technology system. It achieves precise control of generated content through structural constraints, and is the core foundation of its AIGC commercial capabilities.	Gradually implemented and commercialized after 2023	Full-scenario AIGC commercial services including text-to-image, text-to-video, 3D content generation, AI design, and digital humans

1.2 Technical Link Benchmark

The core technical link anchored in this report is: zw Skeleton Network (commercial foundation laid in 1994) → General AIGC Controllable Generation Technology (generalized extension in 2023) → Mainstream AIGC Large Models (large-scale commercial implementation). All analyses are carried out based on this traceable and verifiable complete technology evolution link.

1.3 Benchmark of Key Engineering Facts

1. The 1994 version of *China Grand Character Library* contains 180 sets of national standard secondary character libraries, with each set covering about 6763 Chinese characters. In the conventional TTF font format, it would require 2000-4000MB of storage space, while the standard capacity of CD-ROM at that time was only 650MB. This could only be achieved through the technical solution of "Chinese character skeleton node data storage + real-time rendering and generation via zw Skeleton Network", completing the world's first industrial verification of skeleton-constrained controllable generation.
2. As early as 1994, the zw Skeleton Network had achieved cross-platform 2D/3D rendering implementation, covering multiple systems including PC, MAC and SGI. It treated Chinese characters as special image processing: first generating skeleton data through PC, then importing it into MAC/SGI systems to complete rendering. This verified that the technology is not limited to Chinese character generation, but can be universally applied to all types of 2D/3D image processing scenarios.
3. Up to now, the structural consistency capabilities of the world's mainstream AIGC image and video models are all implemented based on the skeleton-constrained controllable generation technology system, which is highly homologous to the core technical paradigm of the zw Skeleton Network.

II. In-Depth Full-Dimensional Analysis of Academic Homology

This chapter completes the full-link general comparison and analysis from 7 core dimensions: mathematical essence, architectural logic, engineering process, input and output, application scenarios, technical homology, and substantial similarity.

2.1 Homology Analysis of Mathematical Essence

The two are completely isomorphic in mathematical essence, both belonging to conditional generation models, with highly unified core logic:

- **zw Skeleton Network:** By constructing an explicit geometric structure constraint function, it takes the skeleton nodes (center line/contour line nodes) of Chinese characters as constraint conditions, and constrains the high-dimensional random Chinese character generation space to the low-dimensional geometric ordered manifold defined by the skeleton. Its core mathematical expression is:

$$I_{out} = f_{gen}(N_{skel}, \theta_{const}, P_{style}) + \epsilon_{render}$$

where θ_{const} is the structural constraint function, and N_{skel} is the skeleton node input.

- **General AIGC Controllable Generation Technology/Mainstream Large Models:** It adds a control constraint loss term to the loss function of the diffusion model, constructs implicit neural constraints through control map features, and constrains the high-dimensional random image generation space to the low-dimensional sub-manifold consistent with the control map. Its core mathematical expression is:

$$I_{gen} = E_{z_0 \sim p_{data}}[D_{\theta}(z_t, t, c_{text}, c_{control})]$$

where $c_{control}$ is the skeleton/contour control map feature.

Core Conclusion: The two have exactly the same mathematical essence. Both "establish a constraint function through the skeleton structure to constrain the high-dimensional random output to a low-dimensional ordered manifold", solving the core contradiction between randomness and consistency of generative AI. They are 100% homologous at the mathematical level.

2.2 Homology Analysis of Architectural Logic

The two adopt exactly the same three-level architecture design, with a matching degree of more than 95% for core modules and highly similar architectural logic. The specific comparison is as follows:

Architecture Level	Core Modules of zw Skeleton Network	Core Modules of General AIGC Controllable Generation Technology/Mainstream Large Models	Matching Degree
Feature Extraction Layer	Self-developed skeleton feature extraction module, which extracts key nodes, hierarchical relationships and structural features of the target, and encodes them into feature vectors recognizable by the generation	Conditional encoder module, which extracts the spatial structure and contour features of the control map, and encodes them into deep feature vectors recognizable by the diffusion model	98%

	model		
Constraint Fusion Layer	In-depth fusion of skeleton features and generation model, constraining the entire generation process without damaging the native generation capability of the base model, while locking the output structure from deviation	Zero-convolution feature fusion module, where control features are deeply bound to the UNet structure of the diffusion model, freezing the weights of the pre-trained model to preserve the native generation capability while locking the generation structure	95%
Verification and Output Layer	Secondary alignment verification between the generated result and the input skeleton to correct structural deviations, output standardized content that meets the constraints, and support fully automatic batch processing	Structural matching verification between the generated result and the control map to ensure that the main contour/pose is consistent with the input, output image/video content that meets the constraints, and support batch generation	99%

Core Conclusion: The underlying architectures of the two fully follow the core paradigm of "structure first → conditional control → drift suppression → consistency guarantee". The only difference lies in the implementation carrier (the zw Skeleton Network is based on classical symbolism/early connectionism, while general controllable generation technology is based on deep diffusion models). There is no essential innovation in architectural logic, and they have complete academic homology.

2.3 Full-Link Comparison of Engineering Processes

The whole process of engineering implementation of the two is highly matched, with 100% homology in core execution steps. The specific comparison is as follows:

1. **Standard Engineering Process of zw Skeleton Network:** Skeleton node input → skeleton feature extraction and encoding → structural knowledge base constraint fusion → ai_curve intelligent curve rendering → structural consistency verification → compliant content output

2. **Standard Engineering Process of General AIGC Controllable Generation Technology/Mainstream Large Models:** Control map input → control feature extraction and encoding → zero-convolution feature fusion → progressive denoising of diffusion model → structural matching verification → controllable content output

Core Conclusion: The core logic of the engineering processes of the two is completely consistent, both realizing a full closed-loop engineering path of "input constraint - feature fusion - full-process control - result verification". The engineering process of current mainstream AIGC large models completely reuses the engineering paradigm verified and matured by the zw Skeleton Network as early as 1994, with only the upgrade of computing power carrier and model scale, and no innovation in the underlying engineering logic.

2.4 Comparison of Input and Output Dimensions

Comparison Dimension	zw Skeleton Network	General AIGC Controllable Generation Technology/Mainstream Large Models	Core Homology Conclusion
Core Input	Skeleton nodes and contour line drawings of Chinese characters/graphics, with the core being the structural invariants of the target object	Skeleton maps, edge maps and depth maps of images/videos, with the core being the structural features of the target object	The input essence is completely consistent, both being the structural constraint benchmark of the target object
Core Output	Chinese character glyphs, 2D special effect images, 3D rendering models	Images, videos, 3D content that comply with the structural	The output objectives are completely consistent, both

	that comply with the structural constraints of the input skeleton	constraints of the input control map	being structurally precise and controllable generated content without deformation
Core Constraint Objective	Solve the problems of structural deformation, style jump and lack of cross-sample consistency in the batch generation process	Solve the problems of subject distortion, structural out of control and lack of inter-frame consistency in the generation process	The core industry pain points solved are 100% consistent

2.5 Comparison of Scalability of Application Scenarios

The zw Skeleton Network completed multi-scenario engineering implementation as early as 1994, and its application boundary fully covers the core commercial scenarios of current general AIGC controllable generation technology/mainstream large models. The specific comparison is as follows:

Scenario	Implementation Time of zw Skeleton Network	Implementation Time of Mainstream Large Models	Comparison Conclusion
Batch Generation of Chinese Fonts	1994	After 2023	The zw Skeleton Network is the pioneer of this scenario, and current mainstream related applications are scenario extensions
2D Image Processing and Special Effect Generation	1994	After 2023	The technical logic is completely consistent, only the generalization of generated

			objects
3D Content Rendering and Generation	1994	After 2023	The zw Skeleton Network completed the engineering verification of this scenario 30 years earlier, with a homologous technical paradigm
Digitization of Ancient Books and Low-Resolution Text Processing	2018	After 2023	The zw Skeleton Network has special technical advantages, and mainstream related capabilities reuse its core logic

Core Conclusion: The controllable generation commercial scenarios of current mainstream AIGC large models are completely within the technical application boundary already verified by the zw Skeleton Network, only the large-scale expansion of scenarios, without innovation in the underlying application paradigm.

2.6 Comprehensive Judgment of Technical Homology and Substantial Similarity

Based on the above full-dimensional analysis, the core conclusions on the technical homology and substantial similarity between the two are as follows:

1. **Technical Homology:** The zw Skeleton Network is the absolute pioneer of the global "skeleton-constrained controllable generation" technology route. General AIGC controllable generation technology is the generalized extension of this technology route in the deep learning era, and mainstream AIGC large models are the large-scale commercial implementation of this technology route. The three have complete "technical homology in a continuous line".
2. **Substantial Similarity:** General AIGC controllable generation technology/mainstream AIGC large models and the zw Skeleton Network are 100% homologous in core design ideas, mathematical essence, underlying architecture, control mechanism, engineering process and industry pain points solved, with highly similar generation effects, constituting substantial similarity at the level of computer software and technical solutions.

III. Core Dual Judgment: Duck Test + Academic Theoretical Test

3.1 First Judgment: Duck Test

3.1.1 Judgment Criteria

The core criterion of the classic Duck Test is: "If it looks like a duck, swims like a duck, and quacks like a duck, then it probably is a duck". This criterion is fully consistent with the core judgment standard of "access + substantial similarity" in the judicial practice of copyright infringement in China.

3.1.2 Itemized Judgment

1. Judgment of Substantial Similarity (Like a Duck)

The controllable generation technology integrated in mainstream AIGC large models is completely consistent with the zw Skeleton Network in core design ideas, mathematical models, underlying architecture, engineering process and application effects. The core function of both is "to realize structural control and consistency guarantee of generated content through skeleton constraints". From the three dimensions of user perception, technical implementation and engineering effect, they fully meet the substantial similarity standard of "looks like, uses like, and runs like".

2. Judgment of Access Possibility (With Access Channels)

The zw Skeleton Network was commercially released in 1994, its core academic paper was published in 1999, and since 2008, it has fully open-sourced the core code, technical principles, algorithm documents and application cases through global public channels such as Cnblogs and GitHub. It has been continuously open to the public for more than ten years before the large-scale application of general controllable generation technology in 2023. The R&D teams of the world's mainstream AI enterprises have complete technical access paths, the access possibility has high probability, and there is no reasonable defense space.

3. Exclusion of the Possibility of Coincidental Independent Development (Not Independently Developed)

In the field of computer technology, the probability of independently developing a technical solution that is completely consistent from the underlying mathematical logic, architecture design to the whole engineering process for the same core industry pain point is infinitely close to zero. Moreover, the zw Skeleton Network has fully disclosed all technical details more than ten years in advance, so there is no reasonable possibility of "working behind closed doors and happening to have the same idea" for general controllable generation technology and related technologies of mainstream large models.

3.1.3 Final Judgment of the Duck Test

The skeleton-constrained controllable generation technology integrated in mainstream AIGC large models constitutes substantial similarity with the zw Skeleton Network technology, fully complying with the core judgment criteria of the Duck Test. Its underlying technical paradigm is derived from the zw Skeleton Network, and the technical source is not standardizedly declared, which violates the open source industry practices and the principle of commercial integrity.

3.2 Second Judgment: Academic Theoretical Test

3.2.1 Judgment Criteria

From the four core dimensions of "ideological origin - paradigm innovation - engineering verification - academic inheritance" at the academic level, the professional judgment of academic homology and innovation is completed. All conclusions are based on publicly traceable academic achievements and engineering facts.

3.2.2 Itemized Judgment

1. Ideological Origin Dimension

The core idea of the zw Skeleton Network is derived from the frame structure theory of "establishing the skeleton first, then filling the flesh and blood" in Chinese calligraphy. It completed digital and engineering transformation in 1990, and is the world's first system that fully engineered the complete process of "structural constraint → content generation", forming the original Eastern, structure-driven AI technical ideology. The controllable generation ideas of general controllable generation technology and mainstream AIGC large models fully inherit this core paradigm, without original academic ideological innovation.

2. Paradigm Innovation Dimension

The zw Skeleton Network created the controllable generation academic paradigm of "rigid constraint of skeleton contour + in-depth fusion of generation model", which fundamentally solved the core industry pain points of generative AI: "strong randomness, structural deformation, and lack of consistency". This paradigm is still the core underlying logic of global AIGC controllable generation to this day. General controllable generation technology is only the generalized extension of this paradigm in the deep learning era, and mainstream AIGC large models are only the engineering application of this paradigm, neither of which has broken through the academic paradigm boundary established by the zw Skeleton Network.

3. Engineering Verification Dimension

As early as 1994, the zw Skeleton Network completed large-scale commercial verification of 180 sets of national standard secondary character libraries and more than 670,000 Chinese characters, broke through the physical storage limit of 650M CD-ROM, and completed engineering implementation in 2D/3D multi-scenarios. Its engineering logic of "constraint function + real-time generation + consistency

verification" is still the core engineering standard of controllable generation technology to this day. The engineering paths of general controllable generation technology and mainstream AIGC large models completely reuse this engineering logic, with only the upgrade of computing power carrier and model scale, and no academic innovation in the underlying engineering logic.

4. Academic Inheritance Dimension

The technical system of the zw Skeleton Network was formally established through the core paper *Artificial Intelligence and Chinese Font Design* in 1999, and was listed as one of the three basic modeling theories for Chinese character libraries in the industry. Since 2008, it has continuously published academic achievements through public channels, forming a complete academic communication link. The related technology R&D of general controllable generation technology and mainstream AIGC large models is all within the public communication cycle of the academic achievements of the zw Skeleton Network, constituting an inheritance and reference relationship at the academic level, and the technical source is not declared in accordance with academic norms.

3.2.3 Final Judgment of the Academic Theoretical Test

The zw Skeleton Network is the pioneer and academic founder of the global controllable generation AIGC academic paradigm. General AIGC controllable generation technology is the extension of this academic paradigm in the deep learning era, and mainstream AIGC large models are the large-scale appliers of this academic paradigm. The three have complete "inseparable academic homology". The related technologies of mainstream large models have not standardizedly declared their academic sources, which violates the basic norms and integrity principles of academic research.

IV. Legal Risk and Compliance Analysis of Both Parties

4.1 Legal Right Basis of Party zw

1. **Copyright Right Basis:** He Haiqun enjoys complete copyright over the original technical papers, algorithm documents, software codes, related font works of *China Grand Character Library*, and engineering achievements of the zw Skeleton Network, which are protected by the *Copyright Law of the People's Republic of China* from the date of completion of creation. The complete timeline of commercial release in 1994, paper publication in 1999, and continuous open source since 2008 constitutes an unbreakable chain of evidence of ownership.

2. **Prior Right to Technology:** The core technology of the zw Skeleton Network was commercially implemented in 1994, 29 years earlier than the large-scale application of general controllable generation technology, and more than 30 years earlier than the implementation of related controllable generation capabilities of

mainstream large models. It has absolute technical priority, constitutes prior art in the sense of the *Patent Law*, can be used to counter third-party related patent claims, and enjoys the legal right of prior use.

3. **Legal Rights Related to Open Source:** The core technical solution of the zw Skeleton Network was open-sourced and released through global public channels. Although no single open source license is clearly defined, open source industry practices and the *Anti-Unfair Competition Law* clearly require that the source of technology should be standardizedly declared for commercial use of open source technology. The Apache 2.0 License, which is commonly used by mainstream open source controllable generation technologies, also clearly requires that the copyright notice be retained and changes be declared for modifications. The act of not standardizedly declaring the technical source has violated the open source rules, and Party zw enjoys the legal right to pursue liability.

4.2 Core Compliance Risk Points of AIGC Large Model Manufacturers

1. **Copyright Infringement Risk:** The controllable generation technology integrated in mainstream AIGC large models constitutes substantial similarity with the zw Skeleton Network, and the R&D team has access possibility, which fully meets the core elements of "access + substantial similarity" for copyright infringement determination in China. If Party zw claims the copyright of the original expression of the specific algorithm implementation, constraint function design, engineering process architecture, and code in the technical solution, the manufacturers will face direct infringement risks, and the large-scale dissemination of their downstream commercial scenarios will further expand the infringement results and compensation liability.

2. **Unfair Competition Risk:** As leading players in the industry, AI enterprises carry out large-scale commercial profit-making activities based on controllable generation technology with undeclared technical sources, use the decades of prior technical achievements of Party zw to obtain market competitive advantages, violate the principle of good faith and business ethics stipulated in the *Anti-Unfair Competition Law*, and constitute a "free-riding" unfair competition behavior, facing multiple risks of administrative penalty, civil compensation and damage to brand reputation.

3. **Open Source Compliance Risk:** The core technical paradigm of general controllable generation technology is derived from the prior open source achievements of the zw Skeleton Network, but the technical source is not standardizedly declared in accordance with open source rules, which itself has open source compliance defects. Mainstream AIGC large models carrying out commercial development based on this non-compliant open source technology will face joint accountability from the open source community, and at the same time affect their compliance in overseas markets, causing potential obstacles to their global business layout.

4. **Chain Risks of Commercial Rights Protection:** If Party zw adopts the rights

protection strategy of "encircling the upstream from the downstream", and prioritizes launching rights protection against the terminal commercial customers of large models (listed companies, state-owned enterprises and central enterprises, leading brands with annual revenue of more than 1 billion yuan), it will directly affect the commercial reputation of the products and customer trust, force the manufacturers to face the compliance issues of the technical source directly, and face large-scale commercial claims and fluctuations in business operation.

4.3 Core Legal Dispute Focuses

1. **The Boundary of the Idea-Expression Dichotomy:** Manufacturers may plead that "structure-constrained generation belongs to the category of ideas and is not protected by the copyright law"; however, Party zw can claim that the specific algorithm implementation, constraint function design, engineering process architecture, code implementation, and non-general original details in its technical solution belong to the category of "expression" protected by the copyright law, rather than abstract ideas.
2. **Scope of Determination of Substantial Similarity:** The core dispute is whether the infringement determination is limited to literal similarity at the code level, or extends to non-literal substantial similarity of algorithm architecture, design ideas, engineering processes, and module design. In the current judicial practice in China, the determination of computer software infringement has gradually recognized the infringement determination standard of "non-code level substantial similarity of architecture, logic and process".
3. **Boundary of Commercial Use of Open Source Technology:** The core dispute is whether the technical solution that has been open-sourced and made public can be used commercially for free. Judicial practice has clearly stated that even if the technical solution has been open-sourced, commercial use without standardizedly declaring the technical source and use beyond the scope of the open source license can still constitute unfair competition and copyright infringement.

V. Two-Way Recommendations at the Academic Level

5.1 Academic Recommendations for Party zw

1. **Improve the Systematic Construction of Academic Achievements:** Systematically sort out the technical R&D data, papers, engineering achievements, open source records, and patent documents from 1990 to the present, form a complete academic genealogy archive, publish review papers in core journals and top academic conferences in the field of computer vision and AI, formally establish the academic foundational status of the zw Skeleton Network in the field of global controllable generation AIGC, and fill the gaps in academic traceability.
2. **Promote the Authoritative Recognition of the Academic Community:** Jointly issue expert opinion letters on academic homology with authoritative experts in the

fields of computer vision, AIGC, and Chinese character libraries, organize national academic seminars to conduct authoritative recognition of the academic inheritance relationship between the zw Skeleton Network and general controllable generation technology, form industry-recognized academic conclusions, and provide core academic support for the justification of the technology.

3. **Deepen Technical Research and Consolidate Academic First-Mover**

Advantage: Based on the core paradigm of the zw Skeleton Network, further deepen the research on the LCE Logical Consistency Engine and AGM Micro-Weight Graph, launch an upgraded technical solution adapted to the era of large models for the current industry pain points of large model hallucination and lack of long video consistency, form a complete closed loop from academic foundation to technology leadership, and consolidate the academic first-mover advantage of China's native AI technology.

4. **Standardize the Intellectual Property Protection of Open Source**

Achievements: Carry out copyright registration and blockchain deposit certification for core technical papers, algorithm documents, and open source codes, clarify the boundary between open source licenses and commercial use, not only ensure the open dissemination of academic achievements, but also lock the legal authorization boundary of commercial use, and avoid unauthorized commercial use of technical achievements.

5.2 Academic Recommendations for AIGC Large Model Manufacturers

1. **Complete Academic Traceability and Standardized Declaration of Technical Sources**

Sources: Organize the internal R&D team to conduct a comprehensive traceability of the academic sources of controllable generation technology, formally confirm the academic foundational status of the zw Skeleton Network, standardizedly declare the technical source and academic inheritance relationship of the zw Skeleton Network in relevant technical papers, product white papers, and open source documents, abide by academic norms and open source community practices, and avoid academic misconduct risks.

2. **Carry Out Two-Way Academic Cooperation and Joint Research:** Carry out in-depth academic cooperation with the team of He Haiqun (Zhiwang AI), jointly establish a joint laboratory for controllable generation AI, based on the core paradigm of the zw Skeleton Network, jointly research and develop the next generation of controllable generation technology adapted to large models, solve core pain points such as large model logical hallucination and long video consistency, and realize two-way empowerment at the academic level.

3. **Improve the Academic Compliance System for Internal R&D:** Establish an academic traceability and compliance review mechanism for AI technology R&D, conduct comprehensive technical source traceability and intellectual property review for all integrated open source technologies and third-party technical solutions, standardizedly declare technical sources, avoid similar academic compliance risks,

and ensure the compliance and rigor of the R&D system.

4. **Promote the Global Dissemination of China's Native AI Technology:** Relying on the global academic influence of the enterprise, promote the global dissemination of the China's native AI technology paradigm represented by the zw Skeleton Network, and publicize the first-mover achievements of China's original controllable generation technology in top academic conferences and international industry forums, so as to enhance the global academic discourse power of China's AI technology.

VI. Two-Way Recommendations at the Commercial Level

6.1 Commercial Recommendations for Party zw

1. **Adopt the Rights Protection Strategy of "Encircling the Upstream from the Downstream" to Prioritize Commercial Payment Collection:** Prioritize batch fixation of commercial infringement evidence for terminal commercial customers of large models (A-share listed companies, state-owned enterprises and central enterprises, leading brands with annual revenue of more than 1 billion yuan), send lawyer's letters, prioritize realizing payment collection through non-litigation settlement, accumulate judicial precedents, strengthen the core bargaining chip in negotiations with manufacturers, and realize the realization of the commercial value of the technology in the shortest time.
2. **Build a Hierarchical Authorization System to Maximize Commercial Value:** Build a standardized hierarchical technology authorization system, launch a long-term cooperation plan of annual technology authorization + joint R&D for leading AI enterprises; launch low-cost standardized annual authorization packages for small and medium-sized enterprises; keep open source and free for individual non-commercial scenarios, which not only ensures the inclusive dissemination of technology, but also realizes the sustained and stable realization of commercial value.
3. **Prioritize Commercial Settlement to Achieve Win-Win Results for the Industry:** On the basis of a complete evidence chain, prioritize commercial negotiations with major AI enterprises, launch diversified settlement solutions such as technology authorization, IP cooperation and joint R&D, avoid the time and energy costs of long-term litigation, rely on the platform, traffic and resources of leading enterprises to realize large-scale commercial implementation of the technology, and reach a win-win commercial result.
4. **Adopt the Full-Risk Agency Model to Launch Rights Protection with Zero Cost:** Sign a full-risk agency contract with a lawyer team with experience in handling intellectual property rights protection of listed companies and font infringement cases, where the lawyer advances all costs of evidence collection, notarization, litigation and travel expenses, and the revenue is shared according to the agreed proportion after payment is received, to launch rights protection actions with zero upfront cost and zero risk, and ensure the professionalism and execution of rights protection.

5. **Build a Technology IP-Based Ecosystem to Expand Diversified Commercial Scenarios:** Based on the core technology of the zw Skeleton Network, cooperate with enterprises in the fields of font design, AIGC, cultural and creative industries, 3D modeling, and digitization of ancient books, launch co-branded products and customized solutions, build the "Zhiwang AI" technology IP, form a diversified commercial profit model, and amplify the long-term commercial value of the technology.

6.2 Commercial Recommendations for AIGC Large Model Manufacturers

1. **Take the Initiative to Carry Out Compliance Negotiations to Resolve Legal Risks at One Time:** Take the initiative to connect with the team of He Haiqun (Zhiwang AI), carry out technical compliance negotiations, obtain the global legal commercial use authorization of the zw Skeleton Network technology by paying technology licensing fees and signing long-term cooperation agreements, resolve the legal risks of copyright infringement and unfair competition at one time, and avoid the negative impact of large-scale litigation on product commercial operation, brand reputation and capital market valuation.
2. **Carry Out In-Depth Technical Cooperation to Enhance the Core Competitiveness of Products:** Carry out in-depth technical cooperation with the Zhiwang AI team, deeply integrate the core technology of the zw Skeleton Network and the LCE Logical Consistency Engine into the self-owned large model, optimize the controllable generation capability, logical consistency and long video generation effect of the large model, solve the core user pain points such as current large model hallucination and structural deformation, and greatly improve the technical barrier and market competitiveness of the product.
3. **Jointly Build a Benchmark of China's Native AI Technology to Enhance Brand Value:** Jointly release a white paper on China's native controllable generation AI technology with the Zhiwang AI team, jointly hold industry summits, publicize the development history of China's technology from the zw Skeleton Network to modern AIGC controllable generation, build a benchmark of China's original AI technology, enhance the brand image and industry discourse power of the enterprise, and avoid technical compliance and intellectual property risks in overseas markets.
4. **Improve the Full-Process Compliance System for Commercial Use of Open Source Technology:** Conduct a comprehensive compliance review of open source technologies integrated in all products of the company, establish a full-process compliance management system of "technology traceability - authorization review - source declaration - commercial monitoring", clarify the boundary of commercial use of open source technologies, standardize the declaration of technical sources, avoid similar compliance risks, and ensure the stable and compliant operation of the company's global business.
5. **Establish a Pre-Warning Mechanism for Intellectual Property Risks:** For the core technology R&D in the AI field, carry out prior technology traceability and patent

risk investigation in advance, identify potential intellectual property risks, proactively resolve risks through technical cooperation, patent cross-licensing, advance authorization and other methods, avoid falling into large-scale intellectual property litigation, and ensure the sustained and stable development of core business.

VII. Comprehensive Conclusion

1. **Academic Homology Conclusion:** The zw Skeleton Network is the pioneer and academic founder of the global "skeleton-constrained controllable generation" technology paradigm. General AIGC controllable generation technology is the generalized extension of this paradigm in the deep learning era, and mainstream AIGC large models are the large-scale engineering application of this paradigm. The three have complete "inseparable academic homology", and the current mainstream AIGC related technologies constitute substantial similarity at the technical level with the zw Skeleton Network.

2. **Legal Compliance Conclusion:** Party zw has a complete prior right basis and an unbreakable chain of ownership evidence. Major AI manufacturers face multiple legal risks such as copyright infringement, unfair competition, and open source compliance. If the two parties fail to reach a settlement, the rights protection actions initiated by Party zw will have a significant impact on the commercial operation and brand reputation of related products.

3. **Development Recommendation Conclusion:** At the academic level, the industry should jointly promote the academic traceability and systematic construction of China's native AI technology, standardize academic declarations, carry out joint research, and jointly enhance the global academic discourse power of China's AI technology; at the commercial level, the right holder and the applier should prioritize resolving legal risks through compliant authorization and in-depth technical cooperation, maximize the technical value, reach a win-win commercial result, and jointly promote the innovation and development of China's original AI technology.

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